

Interlock Narrative
Sulfuric Acid Plant

Lithium Americas Corp.

Project Name: **Thacker Pass Project**



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Change Log

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1. INTRODUCTION

This document defines all Safety Interlocks, Permissives, and On-Off Control Logic for Thacker Pass Sulfuric Acid Plant (SA1). It ensures clarity for operations, maintenance, and safety compliance, and serves as a reference for audits and troubleshooting.

Each interlock includes a description, all the conditions / initiators of interlock logic and final output which affects an equipment operation or DCS (Distributed Control System) control point.

This document must be read in conjunction with the P&IDs, the control loop narrative document No. 1500-IN-CTN-0000-EXP-5001.

- Each interlock is applied to only one final element, device, or equipment, but different interlocks could have common initiators.
- Interlock command to motors is not hard-wired. It will be part of the motor start/stop signal and will be communicated through Modbus TCP/IP communication network to motor smart MCC.
- Interlocks are used either to trip a device or equipment immediately (Trip interlock) or to control automatic operation (start/stop) of a motor or sequence (Operational interlock).
- Trip interlock initiates an alarm and should be acknowledged by operators.
- Trip Interlock status shall be back to normal only if initiator conditions are back to normal and reset by an authorized operator.
- Interlock alarm status change, acknowledgement and reset procedure shall follow general guidelines of ISA-18.2 (management of alarm systems for process industry).
- Operational interlocks do not need to be acknowledged and/or reset by an operator. Operational interlocks do not initiate an alarm unless specified otherwise.
- All physical initiators (transmitter, switch) of trip interlock have soft Maintenance Bypass (Override) Switch (MOS) on DCS console. Only authorized operators have access to override (bypass) an initiator.
- A continuous red blinking alarm stays ON at the top of the related graphic page on DCS HMI (Human Machine Interface) when an initiator is bypassed by an operator (text is "BYPASS").
- All instruments, equipment motors, etc. can become "Out of Service" on DCS HMI. Only authorized operators have access to put equipment out of service.

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- A continuous red blinking alarm stays ON at the top of the related graphic page on HMI when any equipment becomes out of service by an operator (text is “OUT OF SERVICE”).
- An authorized operator may override (bypass) an operational interlock on the equipment DCS face plate.
- Interlocks for motors only describe conditions which shall automatically stop or start a motor. Detailed information of motor control including DCS on/off capability, auto/manual control options, and DCS interface and status signals will be described in loop narrative, document No. 1500-IN-CTN-0000-EXP-5001.
- Solenoid valves are normally energized in accordance to fail safe operation principles. An interlock effect shall de-energize a solenoid valve.
- When an interlock trips a solenoid on a modulating control valve, interlock shall put the corresponding controller on manual mode with a safe output consistent with fail safe position of control valve (output of 0% for fail-closed valve and output of 100% for fail-open valve).
- Input of physical switch to DCS is 24 VDC (ON status) in normal conditions (normally closed contact) to follow fail safe operation principles. However, open and closed position switches of valves are normally open when not activated (i.e. when valve is not open and closed positions).
- Initiator set points specified in this document are finalized during commissioning.
- TBD denotes: To Be Determined during commissioning.

2. GENERAL PRINCIPLES FOR INTERLOCKS

Except for hard wired safety signals, all individual inputs to Interlocks will have a Maintenance Override switch (software driven from Interlock overview graphic) which will allow each input to be overridden in engineering mode.

Each interlock which has been activated will have to be reset manually by the operator once its conditions have disappeared except where specified otherwise here after. This philosophy shall be reviewed during detail engineering.

When an Interlock signal requires the shutdown from a PID controller, the controller, whether in Manual or Auto mode, will force its output to the safe state of the process. This Interlock signal shall have the same bypass possibilities as described above.

3. GENERAL PRINCIPLES FOR A CONTROLLER

Unless otherwise described hereafter, the control room operator will be able to choose, for a PID controller, two or three different modes of operation accessible through the controller “faceplate” as follows:

- either “Manual”

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- or “Automatic”
- or “Cascade” for a slave controller (when applicable only)

When selecting the mode “Manual,” the control room operator can operate the control valve related to the control loop by manually setting the output percentage of the controller.

When selecting the “Automatic” mode, the controller will automatically vary its output in proportion to preset parameters until its “Process Value” equals its “Setpoint”.

In “Cascade” Mode, a controller will work in a similar manner to “Automatic” mode except it will receive its “Setpoint” from the output of another Controller.

When an Interlock signal requires the shutdown from the controller, the controller, whether in Manual or Auto mode, will force its output to the safe state of the process.

4. GENERAL PRINCIPLES FOR AN AUTOMATIC ON/OFF VALVE

Unless otherwise described hereafter, the control room operator will be able to choose two different modes of operation accessible through the on/off valve (XV) “faceplate,” as follows:

- either “Manual”
- or “Automatic”

In Manual mode, the on/off valve (XV) will be operated by the control room operator from the faceplate of the on/off valve (XV) using the OPEN or CLOSE buttons.

In Automatic mode, the on/off valve (XV) will be set in its normal operating position and will be controlled by the system and by the automatic sequences.

In case of interlock, the control system will force the position of the valve according to the logic. The valves will remain in the interlocked position, in manual mode, until the operator has reset the interlock and selected the Automatic mode (unless otherwise specified here after). According to the XV valve position the valve representation will dynamically change from GREEN (open - ZSO) to GRAY (closed - ZSC).

5. ABBREVIATIONS

The following is a list of commonly used abbreviations in this document:

Abbreviation	Description
ACC	Air Cooled Condenser
BFW	Boiler Feedwater
CIP	Cold Interpass
COMB	Combination
COMP	Compressor
CONC	Concentration
COND	Condensate
CONV	Converter
CW	Cooling Water
CLWR	Cooling Water Return
CLWS	Cooling Water Supply

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DISCH	Discharge
DT	Drying Tower
ECON	Economizer
EXCH	Exchanger
FAT	Final Absorbing Tower
HIP	Hot Interpass
IPAT	Interpass Absorbing Tower
PRESS	Pressure
PGS	Process Gas
RECIRC	Recirculation
SA	Strong Acid
TEMP	Temperature
TW	Treated Water
TGS	Turbo Generator System
WHB	Waste Heat Boiler
WT	Water Treatment

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6. UNITS OF MEASUREMENT

MEASURED VARIABLE / UNIT DESCRIPTION (METRIC)	USC UNITS	SI UNITS
ACIDITY/BASICITY	pH	pH
ACCELERATION	ft/sec ² (RMS)	m/s ²
AREA		
Square inch (Square millimeter)	in ²	mm ²
Square foot (Square meter)	ft ²	m ²
CONCENTRATION		
Parts per million	ppm	ppm
CONDUCTIVITY		
Micro Siemens per centimeter	mmho/cm	μS/cm
DENSITY		
Percentage	%	%
Pound per cubic foot (kilogram per cubic meter)	lb/ft ³	kg/m ³
Specific gravity – unitless	sg	sg
ELECTROMOTIVE FORCE (VOLTAGE)		
Volts	V	V
Volts direct current	VDC	VDC
Volts alternating current	VAC	VAC
ELECTRIC CURRENT		
Ampere or amp	A	A
ENERGY		
Horsepower hour (kilowatt hour)	hp-h	kWh

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British thermal unit (Mega Joule)	BTU	MJ
FLOW RATE, VELOCITY		
Feet per second (meters per second)	ft/s	m/s
FLOW, VOLUMETRIC		
Gases (million standard cubic feet per day)	MM SCFD	m ³ /s
Vapors (million standard cubic feet per day)	MM SCFD	m ³ /s
Steam (pounds per hour)	lb/hr	kg/s
FLOW, MASS		
Liquids (gallons per minute)	GPM @Actual	m ³ /s
Liquids (pounds per hour)	lb/hr	kg/s

Standard Temperature & Pressure: STP: 32° F, 29.92 in Hg

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7. SAFETY INTERLOCK LIST

INTERLOCK	DESCRIPTION
I-1	SHUT DOWN MAIN COMPRESSOR
I-4	CLOSE MAIN COMPRESSOR DISCHARGE VALVE (1540-HV-4080)
I-7	STOP SULFUR FEED PUMPS AND CLOSE SULFUR FEED VALVE
I-8	SHUT DOWN STARTUP BURNER
I-9	ISOLATE DRYING/INTERPASS TOWER DILUTION WATER
I-10	ISOLATE FINAL TOWER DILUTION WATER
I-13	CLOSE COMBINATION PUMP TANK LEVEL CONTROL VALVE (1520-LCV-6040A)
I-14	ISOLATE ACID TO PRODUCT AREA
I-15	CLOSE DILUTION PUMP TANK LEVEL CONTROL VALVE (1520-LCV-6240)
I-16	ISOLATE PRODUCT ACID DILUTION WATER
I-21	SHUT DOWN HP BOILER FEEDWATER PUMPS (1540-PP-017, 1540-PP-018)
I-23	AUTOSTART STANDBY HP BOILER FEEDWATER PUMPS (1540-PP-017, 1540-PP-018)
I-60	SHUT DOWN TAIL GAS SCRUBBER CIRCULATION PUMPS (1530-PP-001 & 1530-PP-002) AND CLOSE SCRUBBER EFFLUENT VALVE (1530-DCV-0006)
I-61	CLOSE TAIL GAS SCRUBBER MAKE-UP WATER VALVE (1530-LCV-0013)
I-62	CLOSE CAUSTIC SUPPLY CONTROL VALVE (1530-ACV-0002)
I-65	SHUT DOWN OXIDATION AIR BLOWERS (1530-GB-001 AND 1530-GB-002)
I-83	OPEN TREATED WATER BYPASS VALVE 1520-XV-6742, CLOSE WATER BLOCK VALVES (1520-XV-6741 & 1520-XV-6738), AND SWITCH 1520-TIC-6731 TO MANUAL WITH OUTPUT 0 TO CLOSE ACID VALVE 1520-TCV-6731
I-100	LIGHT ALARM BEACON 1510-XA-1118A, SET 1510-LIC-1007 TO MANUAL AND OUTPUT TO 0 TO CLOSE SULFUR FEED VALVE 1510-LCV-1007
I-101	STOP LIQUID SULFUR UNLOADING PUMPS 1510-PP-001 AND 1510-PP-002
I-110	STOP FIN FAN COOLING WATER PUMPS 1520-PP-008, 1520-PP-009, 1520-PP-010
I-111	STOP ACID PLANT COOLING TOWER PUMPS 1550-PP-001, 1550-PP-002
I-112	STOP TREATED WATER EXPORT PUMPS 1560-PP-006, 1560-PP-007, 1560-PP-010 AND 1560-PP-011
I-115	START BYPASS CONDENSATE PUMP 1560-PP-019 OR 1560-PP-020
I-121	OPEN OFF-SPEC CONDENSATE RETURN VALVE 1560-XV-7513, CLOSE CONDENSATE RETURN TO BFW PREHEATER VALVE 1560-XV-7514
I-123	SWITCH 1560-TCV-3603 TO MANUAL MODE WITH OUTPUT TO ZERO AND SWITCH 1560-TCV-3604 TO MANUAL MODE WITH 100% OPEN TO BYPASS THE CONDENSATE TO THE WATER TREATMENT ARE SUMP
I-124	STOP WATER RECOVERY PUMPS (1560-PP-021 AND 1560-PP-022)
I-126	STOP ACC CONDENSATE PUMPS (1560-PP-015 AND 1560-PP-016), STOP BYPASS CONDENSATE PUMPS (1560-PP-019 AND 1560-PP-020)
I-140	CLOSE SULFUR VAPOR SCRUBBER BLOWDOWN VALVE (1510-HV-1512)
I-141	CLOSE MAKE UP WATER VALVE (1510-LCV-1524)
I-150	STOP 98.5% ACID RECIRCULATION PUMPS (1570-PP-005 AND 1570-PP-006)

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I-151	CLOSE ACID LOADING VALVE (1570-HV-9202)
I-152	CLOSE ACID LOADING VALVE (1570-HV-9212)
I-153A	STOP ACID HEATER (1570-ZE-003)
I-154A	STOP ACID HEATER (1570-ZE-004)
I-155	CLOSE LEVEL CONTROL VALVE (1520-LCV-6240), TRUCK UNLOADING ON/OFF VALVE (1570-HV-9011), START-UP ACID STORAGE INLET ON/OFF VALVE (1570-HV-9010) AND STOP START-UP ACID UNLOADING PUMP (1570-PP-007)
I-160	STOP CAUSTIC UNLOADING PUMP (1510-PP-003)
I-161	STOP CAUSTIC FEED PUMPS (1510-PP-007 AND 1510-PP-008)
I-162	START STANDBY CAUSTIC FEED PUMP (1510-PP-007 OR 1510-PP-008)
I-167	STOP SA1 EFFLUENT FEED PUMPS 1520-PP-023 AND 1520-PP-024 AND SA1 EFFLUENT STORAGE TANK AGITATOR 1520-AG-003
I-169	STOP TRANSLOADING EFFLUENT UNLOADING PUMP 1520-PP-021 AND SET 1520-XV-4711 TO MANUAL MODE WITH ZERO OUTPUT
I-170	CLOSE CONTROL VALVE 1520-LCV-4705
I-301	TRIP STEAM TURBO GENERATOR AND SET BYPASS DESUPERHEATER VALVE 1560-PCV-6672 TO MANUAL MODE WITH ZERO OUTPUT
I-303	INITIATE AIR COOLED CONDENSER (ACC) UNIT FREEZE PROTECTION SEQUENCE
I-304	AIR COOLED CONDENSER (ACC) UNIT FREEZE PROTECTION TRIP SEQUENCE
I-310	TRIP AIR COOLED CONDENSER (ACC) FAN MOTORS
I-312	TRIP AIR COOLED CONDENSER (ACC) HOGGING SKID LIQUID RING PUMP (1560-AP-001)
I-313	OPEN AIR-COOLED CONDENSER (ACC) HOGGING SKID SEPARATOR FILLING VALVE (V1560-9329)
I-314	TRIP AIR COOLED CONDENSER (ACC) SKID EJECTOR (1550-BN011 OR 1550-BN012)
I-315	FORCE TO CLOSE ALL AIR SUCTION VALVES (V1560-9340)
I-316	FORCE TO CLOSE ALL MOTIVE STEAM VALVES (V1560-9363, V1560-9366)
I-319	TRIP AIR COOLED CONDENSER (ACC) AIR REMOVAL SYSTEM
I-322	TRIP STEAM TURBO GENERATOR AND CHANGE BYPASS DESUPERHEATER VALVE (1560-PCV-6672) TO AUTO MODE
I-324	STEAM TURBO GENERATOR (STG) BYPASS PROTECTION

8. PERMISSIVES LIST

INTERLOCK	DESCRIPTION
I-2	MAIN COMPRESSOR START PERMISSIVE
I-80	HOT HEAT EXCHANGER BYPASS DAMPERS (1540-TCV-5224A & 1540-TCV-5224B) HEAT UP MODE PERMISSIVE (1540-TIC-5224)
I-81	OPEN HYDROGEN VENTS (1520-XV-5805, 1520-XV-6630, 1520-XV-8423) PERMISSIVE
I-156	PRODUCT ACID TRANSFER PUMPS 1570-PP-001/002 START PERMISSIVE
I-164	SULFUR FEED PUMPS 1510-PP-004 AND 1510-PP-005 START PERMISSIVE
I-171	TRANSLOADING EFFLUENT UNLOADING PUMP 1520-PP-021 START PERMISSIVE

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I-200	20% CAUSTIC SOLUTION EXPORT TO LITHIUM PROCESSING PLANT (ON/OFF VALVE 1510-XV-9530) PERMISSIVE
I-201	TREATED WATER EXPORT PUMPS 1560-PP-006 AND 1560-PP-007 (STANDBY) START PERMISSIVE
I-202	150 PSIG STEAM EXPORT STATION (1560-PCV-6011) LITHIUM PROCESSING PLANT START PERMISSIVE
I-302	PERMISSIVE TO OPEN BYPASS DESUPERHEATER VALVE 1560-PCV-6672
I-307	AIR COOLED CONDENSER (ACC) UNIT FAN START UP PERMISSIVES
I-308	PERMISSIVE FOR AIR COOLED CONDENSER (ACC) START-UP SEQUENCE
I-317	PERMISSIVE FOR OPENING ACC HOLDING SKID AIR SUCTION VALVE(V1560-9340)
I-320	PERMISSIVE FOR OPENING EXHAUST STEAM LINE VACUUM BREAKER (V1560-9308)
I-323	PERMISSIVE FOR PUT FAN ASSEMBLY FUNCTION GROUP IN AUTO

9. ON-OFF CONTROL INTERLOCK LIST

INTERLOCK	DESCRIPTION
I-63A	OPEN RAW WATER TO OXIDATION BLOWERS AIR SPARGER ON/OFF VALVE (1530-XV-0656)
I-63B	CLOSE RAW WATER TO OXIDATION BLOWERS AIR SPARGER ON/OFF VALVE (1530-XV-0656)
I-103	START/STOP LIQUID SULFUR UNLOADING PUMPS 1510-PP-001 AND 1510-PP-002
I-117	START, STOP WATER TREATMENT AREA SUMP PUMPS 1560-PP-012 OR 1560-PP-013
I-119	START BOILER BLOWDOWN SUMP PUMP 1540-PP-001
I-120	STOP BOILER BLOWDOWN SUMP PUMP 1540-PP-001
I-122	OPEN/CLOSE DEAERATOR OVERFLOW VALVE 1540-XV-6843
I-137	START AND STOP ACID PLANT COOLING TOWER FAN (1550-CT-001)
I-142	SULFUR VAPOR SCRUBBER (1510-PK-007) FLUSH SEQUENCE STARTS
I-153B	START/STOP ACID HEATER (1570-ZE-003)
I-154B	START/STOP ACID HEATER (1570-ZE-003)

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10. PURPOSE

This document defines all Safety Interlocks, Permissives, and On-Off Control Logic for Thacker Pass Sulfuric Acid Plant (SA1). It ensures clarity for operations, maintenance, and safety compliance, and serves as a reference for audits and troubleshooting.

11. SCOPE

This document provides a comprehensive description of all logic related to Safety Interlocks, Permissives, and On-Off Control for equipment and process devices.

It specifically includes:

- **Safety Interlocks:** Protective functions designed to prevent hazardous conditions by initiating automatic shutdowns or trips when predefined limits are exceeded.
- **Permissive Logic:** Conditions that must be satisfied before equipment can be safely started or remain in operation, ensuring that the process is in a safe state prior to activation.
- **On-Off Control Logic:** Operational logic for discrete devices such as valves, pumps, and other equipment that operate in a fully open/closed or start/stop mode under normal process conditions.

This document does not cover:

- **Normal Control Strategies:** Continuous control loops and process regulation strategies, which are detailed 1500-PR-SPC-0000-EXP-5003 Process Control Philosophy.

12. DEFINITIONS

- **Acknowledge**
Purpose: Confirms operator awareness of an active alarm or interlock.
Function: Silences audible alarms and visual indicators change from flashing to static but does not clear the fault or remove operational blocks.
- **Reset**
Purpose: Clears an interlock or trip condition after the initiating fault has returned to normal.

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Function: Releases the logic block preventing operation. Visual indicators stay alight until the fault is cleared. Reset does not automatically start equipment.

- **Resume Operation**

Purpose: Returns equipment to its normal operating state after reset.

Function: Sends a command to move equipment back to its predefined normal position.

13. APPLICATION OF ACKNOWLEDGE, RESET, AND RESUME

Logic Type	Acknowledge Required	Reset Type	Resume Operation
Safety Interlock	Yes (Manual)	Automatic (when fault clears)	Not applicable by default; applicable only when explicitly specified in the individual interlock description.
Permissive	N/A	N/A	N/A
On-Off Control	N/A	N/A	N/A

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14. SAFETY INTERLOCKS

I-1, SHUT DOWN MAIN COMPRESSOR:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock effect will shut down the main compressor 1540-GB-001.

CONFIGURATION:

Initiator(s):

Low-Low Waste Heat Boiler Steam Drum Level 1540-LSLL-7440 (2oo3: 1540-LSLL-7440A, 1540-LSLL-7440B, 1540-LSLL-7441)

OR

Common High-High Vibration / Thrust on Main Compressor (1540-VAHH-1060)

OR

Common High-High Bearing (Compressor, Gear Box, Motor) Temperature (1540-TAHH-1020)

OR

Low-Low Lube Oil Pressure of Main Compressor Lube Oil System (1540-PALL-1107)

OR

High-High Lube Oil Temperature (1540-TAHH-1096)

OR

DCS Initiated Shutdown (1540-HS-1071)

OR

Low-Low Acid Flow over Final Absorbing Tower unless bypassed at startup (1520-FSLL-6670)

OR

Low-Low Acid Flow over Interpass Absorbing Tower (1520-FSLL-6770)

OR

CONTROL ROOM Shutdown (1540-ESD-1094)

OR

Low-Low Drying Tower Discharge Pressure (1520-PSLL-5804).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040

1520-PR-PID-0000-EXP-5058

1520-PR-PID-0000-EXP-5066

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1520-PR-PID-0000-EXP-5067
1540-PR-PID-0000-EXP-5074
1520-PR-PID-0000-EXP-5114

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I-4, CLOSE MAIN COMPRESSOR DISCHARGE VALVE (1540-HV-4080):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close main compressor discharge valve 1540-HV-4080.

CONFIGURATION:

Initiator(s):

Low-Low Main Compressor Discharge Pressure (1540-PSLL-4002)
AND
Shutdown of Main Compressor (1540-XS-1077).

Permissive (s):

None.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040

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I-7, STOP SULFUR FEED PUMPS AND CLOSE SULFUR FEED VALVE:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will stop sulfur feed pumps (1510-PP-004/005) and close sulfur feed valve (1540-FCV-2602).

CONFIGURATION:

Initiator(s):

High-High Sulfur Burner Outlet Temperature 1540-TSHH-4220 (2oo3: 1540-TSHH-4220A, 1540-TSHH-4220B, 1540-TSHH-4220C)

OR

Low-Low Waste Heat Boiler Steam Drum Level 1540-LSLL-7440 (2oo3: 1540-LSLL-7440A, 1540-LSLL-7440B, 1540-LSLL-7441)

OR

Shutdown of Main Compressor (1540-XS-1077)

OR

Low-Low Main Compressor Discharge Pressure (1540-PSLL-4002) and Low-Low Differential Pressure across Sulfur Burner and Waste Heat Boiler (1540-PDSLL-4200)

OR

Low-Low Flow to Tail Gas Scrubber Reverse Jet (1530-FSLL-0003) and Low-Low Pressure to Tail Gas Scrubber Reverse Jet (1540-PSLL-0004)

OR

All Tail Gas Scrubber Circulation Pumps Not Running (1530-PP-001/002)

OR

High-High Combination Pump Level (1540-LSHH-6040).

Permissive (s):

NONE

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
1520-PR-PID-0000-EXP-5060
1530-PR-PID-0000-EXP-5100
1530-PR-PID-0000-EXP-5102
1540-PR-PID-0000-EXP-5026
1540-PR-PID-0000-EXP-5040

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1540-PR-PID-0000-EXP-5042
1540-PR-PID-0000-EXP-5074

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I-8, SHUT DOWN STARTUP BURNER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will shut down the Startup Burner via Startup Burner Management System 1540-PK-001.

CONFIGURATION:

Initiator(s):

Low-Low Differential Pressure across Sulfur Burner (1540-PDSLL-4200),
AND
Waste Heat Boiler (1540-PDSLL-4200) For >5 Seconds,
OR
High-High Sulfur Burner Outlet Temperature 1540-TSHH-4220 (2oo3: 1540-TSHH-4220A, 1540-TSHH-4220B, 1540-TSHH-4220C),
OR
Operator initiated shutdown, Remote ESD (1540-XE-2993),
OR
Shutdown of Main Compressor (1540-XS-1077)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5029
1520-PR-PID-0000-EXP-5040

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I-9, ISOLATE DRYING/INTERPASS TOWER DILUTION WATER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close Dilution Valve (1520-ACV-6060), open Bleed Valve (1520-XV-6081) and closes Block Valves (1520-XV-6071 & 1520-XV-6080).

CONFIGURATION:

Initiator(s):

Shutdown of Main Compressor (1540-XS-1077)

OR

Low-Low acid concentration strength on Drying & Interpass Towers Analyzer (1520-ASLL-6060),

OR

Shutdown of Drying & Interpass Towers Acid Pump (1520-PP-001).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1540-PR-PID-0000-EXP-5040

NOTE: When Interlock (I-9) is active, enable the Flow Alarm 1520-FSH-6070.

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I-10, ISOLATE FINAL TOWER DILUTION WATER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close Dilution Valve (1520-ACV-6063), open Bleed Valve (1520-XV-6082) and closes Block Valves (1520-XV-6072 & 1520-XV-6083).

CONFIGURATION:

Initiator(s):

Shutdown of Main Compressor (1540-XS-1077)

OR

Low-Low acid concentration strength on Final Towers Analyzer (1520-ASLL-6063)

OR

Shutdown of Final Towers Acid Pump (1520-PP-002).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1540-PR-PID-0000-EXP-5040

NOTE: When Interlock (I-10) is active, enable the Flow Alarm 1520-FSH-6074.

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I-13, CLOSE COMBINATION PUMP TANK LEVEL CONTROL VALVE (1520-LCV-6040A):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close Dilution Valve (1520-ACV-6040A).

CONFIGURATION:

Initiator(s):

Low-Low Combination Pump Tank Level (1520-LSLL-6040)

OR

Drying Tower/Interpass Tower Acid Pump not running (1520-PP-001)

OR

High-High Dilution Pump Tank Level (1520-LSHH-6240)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1520-PR-PID-0000-EXP-5062
 1520-PR-PID-0000-EXP-5067

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I-14, ISOLATE ACID TO PRODUCT AREA:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock closes Dilution Pump Tank Level Control Valve (1520-LCV-6240), shuts down Product Acid Pump (1520-PP-005), shuts down Startup Acid Unloading Pump (1570-PP-007) and closes Startup Acid Unloading Pump Inlet Valve (1570-HV-9011).

CONFIGURATION:

Initiator(s):

Low-Low Dilution Pump Tank Level (1520-LSLL-6240)

OR

High-High Product Acid Storage Tank 1570-TK-0001 Level (1570-LSHH-8601)

OR

High-High Product Acid Storage Tank 1570-TK-0002 level (1570-LSHH-8604)

OR

High-High Startup Acid Storage Tank (1570-TK-0003) Level (1570-LSHH-9009)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5062
 1520-PR-PID-0000-EXP-5064
 1570-PR-PID-0000-EXP-5186
 1570-PR-PID-0000-EXP-5190

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I-15, CLOSE DILUTION PUMP TANK LEVEL CONTROL VALVE (1520-LCV-6240):

DESCRIPTION:

Upon activation of the below initiator, this interlock closes Dilution Pump Tank Level Control Valve (1520-LCV-6240).

CONFIGURATION:

Initiator(s):

Product Acid Pump Shutdown (1520-PP-005)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5062
 1520-PR-PID-0000-EXP-5064

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I-16, ISOLATE PRODUCT ACID DILUTION WATER:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock closes Inlet Water Valve (1520-ACV-6260), opens Bleed Valve (1520-XV-6081), and closes Block Valves (1520-XV-6280 & 1520-XV-6082).

CONFIGURATION:

Initiator(s):

Low-Low acid concentration strength on Product Acid Analyzer (1520-ASLL-6260)

OR

Shutdown of Drying Tower/Interpass Tower Acid Pump (1520-PP-001)

OR

Shutdown of Product Acid Pump (1520-PP-005)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5060
 1520-PR-PID-0000-EXP-5062

NOTE: When Interlock 16 (I-16) is active, enable the Flow Alarm 1520-FSH-6270.

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I-21, SHUT DOWN HP BOILER FEEDWATER PUMPS (1540-PP-017, 1540-PP-018):

DESCRIPTION:

Upon activation of the below initiator, this interlock shuts down HP Boiler Feedwater Pumps 1540-PP-017 and 1540-PP-018.

CONFIGURATION:

Initiator(s):

Low-Low Deaerator (1540-PV-002) Level (1540-LSLL-6842)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5068
 1540-PR-PID-0000-EXP-5070

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I-23, AUTOSTART STANDBY HP BOILER FEEDWATER PUMPS (1540-PP-017, 1540-PP-018):

DESCRIPTION:

Upon activation of the below initiator, this interlock shuts down HP Boiler Feedwater Pumps 1540-PP-017 and 1540-PP-018.

CONFIGURATION:

Initiator(s):

Low-Low HP Boiler Feedwater Discharge Pressure (1540-PSLL-7000)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5070

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I-60, SHUT DOWN TAIL GAS SCRUBBER CIRCULATION PUMPS (1530-PP-001 & 1530-PP-002) AND CLOSE SCRUBBER EFFLUENT VALVE (1530-DCV-0006):

DESCRIPTION:

Upon activation of the below initiator, this interlock shuts down Tail Gas Scrubber Pumps 1530-PP-001 and 1530-PP-002 and closes Scrubber Effluent Valve 1530-DCV-0006.

CONFIGURATION:

Initiator(s):

Low-Low Tail Gas Scrubber Level (2oo2: 1530-LSLL-0013 and 1530-LSLL-0014)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5100
 1530-PR-PID-0000-EXP-5102

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I-61, CLOSE TAIL GAS SCRUBBER MAKE-UP WATER VALVE (1530-LCV-0013):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock closes Tail Gas Scrubber Make-Up Water Valve 1530-LCV-0013.

CONFIGURATION:

Initiator(s):

High-High Tail Gas Scrubber Level (1530-LSHH-0013)

OR

High-High Tail Gas Scrubber Level (1530-LSHH-0014)

OR

Tail Gas Scrubber Circulation Pumps (1530-PP-001 & 1530-PP-002) not running.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5100
 1530-PR-PID-0000-EXP-5102

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I-62, CLOSE CAUSTIC SUPPLY CONTROL VALVE (1530-ACV-0002):

DESCRIPTION:

Upon activation of the below initiator, this interlock closes Tail Gas Scrubber \ Caustic Supply Control Valve 1530-ACV-0002.

CONFIGURATION:

Initiator(s):

Tail Gas Scrubber Circulation Pumps (1530-PP-001 & 1530-PP-002) not Running.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5100
 1530-PR-PID-0000-EXP-5102

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I-65, SHUT DOWN OXIDATION AIR BLOWERS (1530-GB-001 AND 1530-GB-002):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock shuts down the Oxidation Blowers 1530-GB-001 and 1530-GB-002.

CONFIGURATION:

Initiator(s):

High-High Oxidation Air Temperature to Tail Gas Scrubber (1530-TSHH-0654)

OR

High-High Oxidation Air Discharge Pressure to Tail Gas Scrubber (1530-PSHH-0652)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5106

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I-83, OPEN TREATED WATER BYPASS VALVE 1520-XV-6742, CLOSE WATER BLOCK VALVES (1520-XV-6741 & 1520-XV-6738), AND SWITCH 1520-TIC-6731 TO MANUAL WITH OUTPUT 0 TO CLOSE ACID VALVE 1520-TCV-6731:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock opens water bypass valve (1520-XV-6742), closes water block Valves (1520-XV-6741 and 1520-XV-6738) and switch the controller 1520-TIC-6731 to manual close acid valve 1520-TCV-6731.

CONFIGURATION:

Initiator(s):

High-High BFW Preheater Treated Water Outlet Temperature (1520-TSHH-6731) to Deaerator,
OR

High-High BFW Preheater Treated Water Outlet Conductivity (1520-ASHH-6745) to Deaerator,
OR

High-High BFW Preheater Treated Water Outlet Conductivity (1520-ASHH-6746) to Deaerator

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5067

Resume Normal Operation:

Once both the High-High Conductivity (1520-ASHH-6745 & 1520-ASHH-6746) and High-High Temperature (1520-TSHH-6731) conditions have cleared, press Resume Button (or perform manually) to:

- Close valve 1520-XV-6742,
- Open valves 1520-XV-6741 and 1520-XV-6738,
- Return controller 1520-TIC-6731 to Automatic mode.

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I-100, LIGHT ALARM BEACON 1510-XA-1118A, SET 1510-LIC-1007 TO MANUAL AND OUTPUT TO 0 TO CLOSE SULFUR FEED VALVE 1510-LCV-1007:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will turn on the visual alarm/beacon (1510-XA-1118A) to alert the operators of the high level of unloading pit and stop filling the pit.

Level controller 1510-LIC-1007 will be set to manual to force to close the sulfur feed valve 1510-LCV-1007.

CONFIGURATION:

Initiator(s):

High-High Sulfur Pit Level (1510-LSHH-1007A)

OR

High-High Sulfur Pit Level (1510-LSHH-1007B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010

1510-PR-PID-0000-EXP-5011

Resume Normal Operation:

Once the sulfur pit level returns to normal (below HH setpoint), the DCS will automatically restore control:

- Return controller 1510-LIC-1007 to Automatic mode

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I-101, STOP LIQUID SULFUR UNLOADING PUMPS 1510-PP-001 AND 1510-PP-002:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will stop the liquid sulfur unloading pumps (1510-PP-001 and 1510-PP-002).

CONFIGURATION:

Initiator(s):

High-High Sulfur Storage Tank Level (1510-LSHH-1201)

OR

High-High Sulfur Storage Tank Level (1510-LSHH-1202)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
 1510-PR-PID-0000-EXP-5012

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I-110, STOP FIN FAN COOLING WATER PUMPS 1520-PP-008, 1520-PP-009, 1520-PP-010:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Fin Fan Cooling Pumps (1520-PP-008, 1520-PP-009 and 1520-PP-010).

CONFIGURATION:

Initiator(s):

Low-Low Fin Fan Cooling Water Expansion Tank Level (1520-LSLL-1401)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5114

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I-111, STOP ACID PLANT COOLING TOWER PUMPS 1550-PP-001, 1550-PP-002:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Acid Plant Cooling Tower Pumps 1550-PP-001 and 1550-PP-002.

CONFIGURATION:

Initiator(s):

Low-Low Cooling Tower Basin Level (1550-LSLL-2215)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1550-PR-PID-0000-EXP-5122

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I-112, STOP TREATED WATER EXPORT PUMPS 1560-PP-006, 1560-PP-007, 1560-PP-010 AND 1560-PP-011:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Treated Water Export Pumps 1560-PP-006, 1560-PP-007, 1560-PP-010 and 1560-PP-011.

CONFIGURATION:

Initiator(s):

Low-Low Treated Water Storage Level (1560-LSLL-3401)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5134
 1560-PR-PID-0000-EXP-5138

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I-115, START BYPASS CONDENSATE PUMP 1560-PP-019 OR 1560-PP-020:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start Bypass Condensate Pumps 1560-PP-019 or 1560-PP-020 (standby).

CONFIGURATION:

Initiator(s):

Low-Low Booster Pump Discharge Pressure (1560-PSLL-7524)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176

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I-121, OPEN OFF-SPEC CONDENSATE RETURN VALVE 1560-XV-7513, CLOSE CONDENSATE RETURN TO BFW PREHEATER VALVE 1560-XV-7514:

DESCRIPTION:

Upon activation of the below initiator, this interlock will open the Off Spec Condensate Return On/Off Valve 1560-XV-7513 and closes the Condensate return to the BFW Preheater Valve 1560-XV-7514.

CONFIGURATION:

Initiator(s):

High-High Condensate Conductivity Alarm (1560-ASHH-7512) with one (1) minute delay

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176

Resume Normal Operation:

Once the condensate conductivity (1560-AI-7512) returns to normal (below 1560-ASHH-7512 setpoint), the DCS will automatically restore control:

- Open on/off valve 1560-XV-7514
- Close on/off valve 1560-XV-7513

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I-123, SWITCH 1560-TCV-3603 TO MANUAL MODE WITH OUTPUT TO ZERO AND SWITCH 1560-TCV-3604 TO MANUAL MODE WITH 100% OPEN TO BYPASS THE CONDENSATE TO THE WATER TREATMENT ARE SUMP ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close TCV-3102A and opens TCV-3102B to bypass the condensate to the water treatment sump.

Delays the opening of TCV-3102B to give enough time for the liquid level in the tank to fall below an acceptable level and the temperature to drop down as well.

CONFIGURATION:

Initiator(s):

High-High Liquid Level (1560-LSHH-2901) in Recovery Tank (1560-TK-005)

OR

High-High Temperature on Raw Water header (TIT-3102)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5129
1560-PR-PID-0000-EXP-5136

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I-124, STOP WATER RECOVERY PUMPS (1560-PP-021 AND 1560-PP-022):

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Water Recovery Pumps 1560-PP-021 and 1560-PP-022.

CONFIGURATION:

Initiator(s):

Low-Low Recovery Tank Water Level (1560-LSLL-2901)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5129

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I-126, STOP ACC CONDENSATE PUMPS (1560-PP-015 AND 1560-PP-016), STOP BYPASS CONDENSATE PUMPS (1560-PP-019 AND 1560-PP-020):

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop Condensate Pumps (1560-PP-015 and 1560-PP-016) and Stops Bypass Condensate Pumps (1560-PP-019 and 1560-PP-020).

CONFIGURATION:

Initiator(s):

Low-Low Condensate Tank Water Level (1560-LSLL-9320)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5175
 1560-PR-PID-0000-EXP-5176

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I-140, CLOSE SULFUR VAPOR SCRUBBER BLOWDOWN VALVE (1510-HV-1512) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Sulfur Vapor Scrubber Valve (1510-XV-1512).

CONFIGURATION:

Initiator(s):

Low-Low Sulfur Vapor Scrubber Liquid Level (1510-LSLL-1524)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5015

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I-141, CLOSE MAKE UP WATER VALVE (1510-LCV-1524) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Make up Water Valve (1510-LCV-1524).

CONFIGURATION:

Initiator(s):

High-High Sulfur Vapor Scrubber Liquid Level (1510-LSHH-1524)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5015

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I-150, STOP 98.5% ACID RECIRCULATION PUMPS (1570-PP-005 AND 1570-PP-006):

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Acid Recirculation Pumps 1570-PP-005 and 1570-PP-006.

CONFIGURATION:

Initiator(s):

Low-Low Start-up Acid Storage Tank Level (1570-LSLL-9009)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188
 1570-PR-PID-0000-EXP-5190

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I-151, CLOSE ACID LOADING VALVE (1570-HV-9202) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close the Acid Loading Valve (1570-HV-9202) and shuts down Acid Loading pumps (1570-PP-003 and 1570-PP-004).

CONFIGURATION:

Initiator(s):

Totalizer quantity is greater than Set Point (1570-FQI-9201),
 OR
 Acid loading arm high Level limit switch triggered (1570-LAHH-9203,
 OR
 Arm 1570-ZM-001 not in position (1570-ZA-9204),
 OR
 High-High acid loading area sump level (1570-LSHH-9215),
 OR
 Truck not Grounded (1570-XI-9230),
 OR
 Flow Control Valve actuator fail closed (1570-FCV-9201),
 OR
 Local Emergency Shutdown Push Button Triggered (1570-HS-9202B)
 OR
 Local Emergency Shutdown Push Button Triggered (1570-HS-9212B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186
 1570-PR-PID-0000-EXP-5192

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-152, CLOSE ACID LOADING VALVE (1570-HV-9212) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will close the Acid Loading Valve (1570-HV-9212) and shut down the Acid Loading Pumps (1570-PP-003 and 1570-PP-004).

CONFIGURATION:

Initiator(s):

Totalizer > Set Point (1570-FQI-9211),
 OR
 Acid loading arm high Level limit switch triggered (1570-LAHH-9213,
 OR
 Arm 1570-ZM-001 not in position (1570-ZA-9214),
 OR
 High-High acid loading area sump level (1570-LSHH-9215),
 OR
 Truck not Grounded (1570-XI-9231),
 OR
 Flow Control Valve actuator failure (1570-FCV-9211),
 OR
 Local Emergency Shutdown Push Button Triggered (1570-HS-9202B),
 OR
 Local Emergency Shutdown Push Button Triggered (1570-HS-9212B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186
 1570-PR-PID-0000-EXP-5192

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-153A, STOP ACID HEATER (1570-ZE-003) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Acid Heater 1570-ZE-003.

CONFIGURATION:

Initiator(s):

Low-Low Acid Flow (1570-FSLL-8816)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-154A, STOP ACID HEATER (1570-ZE-004) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Acid Heater 1570-ZE-004.

CONFIGURATION:

Initiator(s):

Low-Low Acid Flow (1570-FSLL-8815)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

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I-155, CLOSE LEVEL CONTROL VALVE (1520-LCV-6240), TRUCK UNLOADING ON/OFF VALVE (1570-HV-9011), START-UP ACID STORAGE INLET ON/OFF VALVE (1570-HV-9010), AND STOP START-UP ACID UNLOADING PUMP (1570-PP-007) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close 1520-LCV-6240, 1570-HV-9011, 1570-HV-9010 and stop start the Acid Unloading Pump 1570-PP-007.

CONFIGURATION:

Initiator(s):

High-High Start-Up Acid Storage Tank (1570-TK-003) Level (1570-LSHH-9009)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5190
 1570-PR-PID-0000-EXP-5192

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I-156, PRODUCT ACID TRANSFER PUMP (1570-PP-001 AND 1570-PP-002):

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Product Acid Transfer Pumps 1570-PP-001 and 1570-PP-002.

CONFIGURATION:

Initiator(s):

N/A

Permissive (s):

Lithium Processing Plant must give permission to start the pumps.

(NEED TO HAVE THE LITHIUM PROCESS PLANT READY SIGNAL TAG NUMBER)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186

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I-160, STOP CAUSTIC UNLOADING PUMP (1510-PP-003) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will stop the Caustic Unloading Pump 1510-PP-003.

CONFIGURATION:

Initiator(s):

High-High Caustic Tank (1510-TK-003) Level (1510-LSHH-9516)

OR

Truck Not Grounded (1570-XI-9540)

OR

Emergency Shutdown Push Button Triggered (1510-HS-9512)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5182

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I-161, STOP CAUSTIC FEED PUMPS (1510-PP-007 AND 1510-PP-008) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Caustic Feed Pump 1510-PP-007 and 1510-PP-008 (Standby).

CONFIGURATION:

Initiator(s):

Low-Low Caustic Tank (1510-TK-003) Level (1510-LSLL-9516)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5182

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I-162, START STANDBY CAUSTIC FEED PUMP (1510-PP-007 OR 1510-PP-008) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Caustic Feed Pump 1510-PP-007 or 1510-PP-008, whichever is on standby.

CONFIGURATION:

Initiator(s):

Low Caustic Supply Header Pressure (1510-PSL-9529)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5195

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I-167, STOP SA1 EFFLUENT FEED PUMPS 1520-PP-023 AND 1520-PP-024 AND SA1 EFFLUENT STORAGE TANK AGITATOR 1520-AG-003 ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop SA1 Effluent Feed Pumps 1520-PP-023 and 1520-PP-024 SA1 Effluent Storage Tank Agitator 1520-AG-003.

CONFIGURATION:

Initiator(s):

Low-Low Effluent Storage Tank Level (1520-LSLL-4705)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-169, STOP TRANSLOADING EFFLUENT UNLOADING PUMP 1520-PP-021 AND SET 1520-XV-4711 TO MANUAL MODE WITH ZERO OUTPUT ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will stop Transloading Effluent Unloading Pump 1520-PP-021 and set 1520-XV-4711 to manual mode.

CONFIGURATION:

Initiator(s):

High-High Effluent Storage Tank Liquid Level (1520-LSHH-4705),
OR
Emergency Shutdown Push Button Triggered (1520-HS-9010B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-170, CLOSE CONTROL VALVE 1520-LCV-4705 ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the control valve 1520-LCV-4705.

CONFIGURATION:

Initiator(s):

Low-Low Effluent Storage Tank pH level (1520-ASHH-4711),

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-301, TRIP STEAM TURBO GENERATOR AND SET BYPASS DESUPERHEATER VALVE 1560-PCV-6672 TO MANUAL MODE WITH ZERO OUTPUT ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will trip the Steam Turbo Generator and Set the Bypass Desuperheater Valve 1560-PCV-6672 to manual close with zero output.

CONFIGURATION:

Initiator(s):

High-High Temperature of Exhaust Stem (2003, 1560-TSHH-9312),
OR
High Temperature of Exhaust Steam (1560-HS-9312) and timer (Note 1)
OR
High-High Pressure of exhaust steam (2003, 1560-PSHH-9303),
OR
High-High Condensate Tank (1560-TK-004) Pressure (1560-PSHH-9323),
OR
High-High Condensate Tank (1560-TK-004) liquid level (2003, 1560-LSHH-9320),
OR
High-High Drain Pot Level (1560-LSHH-9407),
OR
DCS Emergency Trip (1560-XS-6210)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5160
 1560-PR-PID-0000-EXP-5166
 1560-PR-PID-0000-EXP-5175

NOTE:

1. Please refer to the figure below as recommended in 1500-IN-VEN-3410-EXP-1201_Functional Description for Air Cooler Condenser Section 2.3 for timber setting vs temperature.

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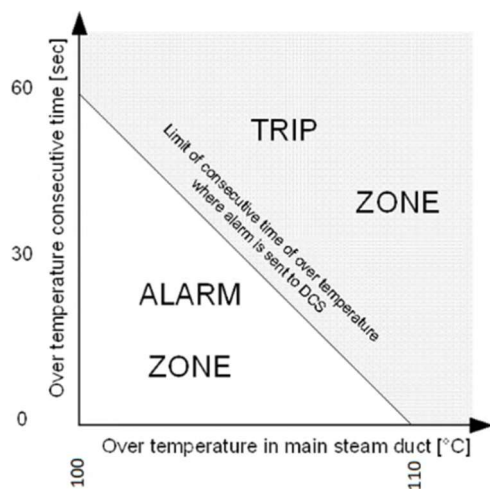


Figure 1 – ACC Trip/Alarm limit

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I-303, INITIATE AIR COOLED CONDENSER (ACC) UNIT FREEZE PROTECTION SEQUENCE:

- I. EVERY 15 MINUTES THE FAN ASSEMBLY OF THE MIX MODULE (1560-AN-001/002/003), IF IT IS RUNNING, WILL BE ALTERNATIVELY SLOWED DOWN BY 20 HZ OR STOPPED IF THE FREQUENCY IS BELOW 15 HZ (MIN VALUE TO START THE FAN) FOR A 10-MINUTE PERIOD TO KEEP THE TUBES WARM.
- II. IF ANY OF THE CONDENSATE TEMPERATURE READING (1560-TE-9326 OR 1560-TE-9328) IS LOWER THAN 25°C (77°F), THEN AN ALARM IS TRIGGERED AT DCS AND THE TURBINE BACKPRESSURE SET-POINT SHALL BE INCREASED 30 MBAR (ADJUSTABLE) EVERY 30 MINUTES UNTIL ALL CONDENSATE TEMPERATURES HAVE RISEN OVER 30°C (86°F). WHEN ALL CONDENSATE TEMPERATURES (1560-TE-9326 OR 1560-TE-9328) ARE OVER 30°C (86°F), THE BP SET-POINT SHALL BE DECREASED 30 MBAR (ADJUSTABLE) EVERY 60 MINUTES UNTIL THE ORIGINAL SET POINT IS REACHED.
- III. TURNING ON HEAT TRACING (1560-HS-9331) FOR INSTRUMENTS, LINES AND VALVES.

CONFIGURATION:

Initiator(s):

Ambient Air Temperature (1560-TE-9300) is lower than - 3°C (26.6°F), for a continuous period of 5 minutes.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

NOTE:

1. Resume Action: The ACC unit freeze protection function will be deactivated when the ambient temperature is higher than 3°C (37.4°F), for a continuous period of 5 minutes.
2. During heating periods, the fan control step shall not be changed.

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I-304, AIR COOLED CONDENSER (ACC) UNIT FREEZE PROTECTION TRIP SEQUENCE:

- I. TRIP STEAM TURBO GENERATOR AND SET BYPASS DESUPERHEATER VALVE 1560-PCV-6672 TO MANUAL MODE WITH ZERO OUTPUT.
- II. SHUTDOWN FAN ASSEMBLY (1560-AN-001, 1560-AN-002 AND 1560-AN-003)
- III. OPEN SKY VENT 1560-PCV-6018

CONFIGURATION:

Initiator(s):

Ambient Air Temperature (1560-TSL-9300) is lower than - 3°C (26.6°F), for a continuous period of 5 minutes,
AND
Low Air-Cooled Condenser Unit flowrate (1560- FSL-6004),
AND
Low Ambient Temperature (1560-TSL-9300)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5166
 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5160

NOTE:

Please refer to the figure below as recommended in 1500-IN-VEN-3410-EXP-1201_Functional Description for Air Cooler Condenser Section 2.3 for minimum flow/temperature limit.

Temperature (°F)	Minimum flow (lb/h)
32	38581
23	52911
14	67242
5	84879
0.9	93697

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I-310, TRIP AIR COOLED CONDENSER (ACC) FAN MOTORS ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will trip the associated Air-Cooled Condenser (ACC) Fan Motor.

CONFIGURATION:

Initiator(s):

1560-AN001

HIGH-HIGH MOTOR VIBRATION (1560-VSHH-9382),

OR

HIGH-HIGH GEAR BOX TEMPERATURE (1596-TSHH-9397 OR 1560-TSHH-9402)

OR

HIGH-HIGH MOTER TEMPERATURE (1560-TSHH-9357, 1560-TSHH-9362, 1560-TSHH-9367, 1560-TSHH-9372, 1560-TSHH-9377)

1560-AN002

HIGH-HIGH MOTOR VIBRATION (1560-VSHH-9383),

OR

HIGH-HIGH GEAR BOX TEMPERATURE (1596-TSHH-9398, 1560-TSHH-9403)

OR

HIGH-HIGH MOTER TEMPERATURE (1560-TSHH-9358, 1560-TSHH-9363, 1560-TSHH-9368, 1560-TSHH-9373, 1560-TSHH-9378)

1560-AN003

HIGH-HIGH MOTOR VIBRATION (1560-VSHH-9384),

OR

HIGH-HIGH GEAR BOX TEMPERATURE (1596-TSHH-9399, 1560-TSHH-9404)

OR

HIGH-HIGH MOTER TEMPERATURE (1560-TSHH-9359, 1560-TSHH-9364, 1560-TSHH-9369, 1560-TSHH-9374, 1560-TSHH-9379)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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I-312, TRIP AIR COOLED CONDENSER (ACC) HOGGING SKID LIQUID RING PUMP (1560-AP-001)
ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the ACC pump 1560-AP-001.

CONFIGURATION:

Initiator(s):

High-High ACC Hogging Skid Separation Tank Liquid Level (1560-LSHH-9333),

OR

Low-Low ACC Hogging Skid Separator Tank (1560-BB-002) Liquid Level (1560-LSLL-9334) for more than 180 seconds.

AND

Closes V1560-9329 upon High-High Separator Tank Level (1560-LSHH-9333)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-313, OPEN AIR-COOLED CONDENSER (ACC) HOGGING SKID SEPARATOR FILLING VALVE (V1560-9329) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will open the ACC Hogging Skid Separation Valve (V1560-9329).

CONFIGURATION:

Initiator(s):

Low-Low ACC Hogging Skid Separator Tank (1560-BB-002) Liquid Level (1560-LSLL-9334) for more than 120 seconds.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-314, TRIP AIR COOLED CONDENSER (ACC) SKID EJECTOR (1550-BN011 OR 1550-BN012) ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the ACC Skid Ejector 1550-BN011 or 1550-BN012.

CONFIGURATION:

Initiator(s):

High-High Motive Steam Temperature (1560-TSHH-9353),

OR

High-High Motive Steam Pressure (1560-PSHH-9352),

OR

Low-Low Gland Steam Condenser Flow (1560-FSLL-7511).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176
 1560-PR-PID-0000-EXP-5177

NOTE:

1560-FSLL-7511 will be triggered when flowrate < 40 t/h (88,185 lbs/h) for 30s (Steam Turbine operation), or < 120 t/h (264,544 lbs/h) for 30s (Bypass operation)

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I-315, FORCE TO CLOSE ALL AIR SUCTION VALVES (V1560-9340) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Air Suction Valve (V1560-9340).

CONFIGURATION:

Initiator(s):

Low-Low Motive Steam Pressure (1560-PSLL-9352)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-316, FORCE TO CLOSE ALL MOTIVE STEAM VALVES (V1560-9363, V1560-9366) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will close the Motive Steam Valves V1560-9340 and V1560-9366).

CONFIGURATION:

Initiator(s):

High-High Motive Steam Pressure (1560-PSHH-9352)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-319, TRIP AIR COOLED CONDENSER (ACC) AIR REMOVAL SYSTEM ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the ACC Air Removal System.

CONFIGURATION:

Initiator(s):

Holding skid Motive Steam Inlet Valves (V1560-9363 or V1560-9366) Fault,

OR

Holding skid air suction valve (V1560-9340) Fault,

OR

Hogging skid air inlet valve (V1560-9324) Fault,

OR

Hogging skid water refill valve (V1560-9329) Fault.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-322, TRIP STEAM TURBO GENERATOR AND CHANGE BYPASS DESUPERHEATER VALVE (1560-PCV-6672) TO AUTO MODE ON:

DESCRIPTION:

Upon activation of any of the below initiators, this interlock will trip the Steam Turbo Generator and change the Bypass Desuperheater Valve (1560-PCV-6672) mode to Auto.

CONFIGURATION:

Initiator(s):

Main Compressor Trip (1540-XS-1077)

OR

High-High Waste Heat Boiler Level (2003, 1540-LSHH-7440)

OR

DCS Emergency Trip (1560-XS-6210)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040
 1540-PR-PID-0000-EXP-5074
 1560-PR-PID-0000-EXP-5160
 1560-PR-PID-0000-EXP-5166

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I-324, STEAM TURBO GENERATOR (STG) BYPASS PROTECTION:

DESCRIPTION:

This interlock will set the Bypass Valve (1560-PCV-6672) into manual mode with zero (0) output. It will only be active when the Steam Turbo Generator Bypass protection signal (1560-XS-6202) is available.

CONFIGURATION:

Initiator(s):

High-High STG last stage pressure (1560-PIA-6317)
OR
High-High STG last stage temperature (1560-TIA-6232)
AND
Bypass Protection Signal (1560-XS-6202)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5160

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15. PERMISSIVES

I-2, MAIN COMPRESSOR START PERMISSIVE

DESCRIPTION:

The interlock permits the Main Compressor to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Main compressor discharge valve 1540-XV-4080 fully open (1540-ZI-4080).

AND

Main compressor inlet guide vanes closed (1540-ZAC-4030).

AND

field-mounted shutdown hand switch disengaged (1540-ESD-1094).

AND

Acid flow over drying tower not low (1540-FSLL-5870).

AND

All hydrogen vents confirmed closed (1540-ZAC-5805, 1540-ZAC-6630, 1540-ZAC-8421).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5040
 1520-PR-PID-0000-EXP-5058
 1520-PR-PID-0000-EXP-5066
 1520-PR-PID-0000-EXP-5084

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I-80, HOT HEAT EXCHANGER BYPASS DAMPERS (1540-TCV-5224A & 1540-TCV-5224B) HEAT UP MODE PERMISSIVE (1540-TIC-5224):

DESCRIPTION:

The interlock permits Heat-up mode (Start-up Bypass mode) from DCS (1540-HS-5224) when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Main Compressor (1540-GB-001) Not Running (1540-XS-1077)
AND
Sulfur Feed Pump (1510-PP-004) Not Running
AND
Standby Sulfur Feed Pump (1510-PP-005) Not Running

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
 1540-PR-PID-0000-EXP-5040
 1540-PR-PID-0000-EXP-5052

Note: Allows manual closure of 1540-TCV-5524A and 1540-TCV-5524B when start up duct is in place.

The valves shall be programmed in the DCS to prevent closing both during normal operation and in manual mode.

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I-81, OPEN HYDROGEN VENTS (1520-XV-5805, 1520-XV-6630, 1520-XV-8423) PERMISSIVE:

DESCRIPTION:

To remove the Hydrogen built up inside the Tower's dome, the On/Off Valves will be in open position during shutdown mode. The interlock permits opening hydrogen vents (1520-XV-5805, 1520-XV-6630, 1520-XV-8423) when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Sulfur Feed Pumps (1510-PP-004 & 1510-PP-005) Not Running 1510-HS-1004A & 1510-HS-1004B),

AND

Main Compressor Speed (1540-XI-1079) less than 200 RPM.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010
 1520-PR-PID-0000-EXP-5058
 1520-PR-PID-0000-EXP-5066
 1520-PR-PID-0000-EXP-5084
 1540-PR-PID-0000-EXP-5040

NOTE: The valves must be in closed position during normal operation which will function as permissive to initiate interlock I-164. This will allow the Sulfur Feed Pumps (1510-PP-004 and 1510-PP-005) start-up.

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I-156, PRODUCT ACID TRANSFER PUMPS 1570-PP-001/002 START PERMISSIVE:

DESCRIPTION:

The interlock permits the Product Acid Transfer Pumps (1570-PP-001/002) to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Process Plant Ready Indication

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5186

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I-164, SULFUR FEED PUMPS 1510-PP-004 AND 1510-PP-005 START PERMISSIVE ON:

DESCRIPTION:

Below initiators must be satisfied as permissive to start the Sulfur Feed Pumps 1510-PP-004 and 1510-PP-005.

CONFIGURATION:

Permissive(s):

Drying Tower Vent Valve Closed (1520-ZAC-5805)
 AND
 Interpass Tower Vent Valve Closed (1520-ZAC-8423)
 AND
 Final Tower Vent Valve Closed (1520-ZAC-6630)
 AND
 Tail Gas Scrubbed pH Normal (1530-AIC-0002)
 AND
 {(Oxidation Blowers Running (1530-GB-001)
 OR
 Oxidation Blowers Running (1530-GB-002))
 AND
 Waste Heat Boiler Not Low or High (1540-HS-7440)
 AND
 Sulfur Burner Outlet Temperature Not Low (1540-TSL-4220)
 AND
 Main Compressor Running (1540-XS-1077)
 AND
 Fin Fan Cooling Water Flow Not Low, 1520-FSL-1413
 AND
 Cooling Tower Water Flow Not Low, 1550-FSL-2218
 AND
 Treated Water Pump (1560-PP-006 OR 1560-PP-007) Running
 AND
 Hot Heat Exchanger Bypass Dampers Not Both Closed (1530-TCV-5224A AND 1530-TCV-5224B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5058
 1520-PR-PID-0000-EXP-5084
 1520-PR-PID-0000-EXP-5066

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1530-PR-PID-0000-EXP-5100
1530-PR-PID-0000-EXP-5106
1540-PR-PID-0000-EXP-5074
1550-PR-PID-0000-EXP-5122
1520-PR-PID-0000-EXP-5114
1560-PR-PID-0000-EXP-5134
1540-PR-PID-0000-EXP-5052

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I-171, TRANSLOADING EFFLUENT UNLOADING PUMP 1520-PP-021 START PERMISSIVE ON:

DESCRIPTION:

The interlock permits the Transloading Effluent Unloading Pumps (1520-PP-021) to start (1520-HS-4702) only when the following condition are satisfied.

CONFIGURATION:

Permissive(s):

Effluent Storage Tank Liquid Level Below High Setpoint (1520-LSH-4705).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1520-PR-PID-0000-EXP-5147

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I-200, 20% CAUSTIC SOLUTION EXPORT TO LITHIUM PROCESSING PLANT (ON/OFF VALVE 1510-XV-9530) PERMISSIVE ON:

DESCRIPTION:

The interlock permits 20% caustic solution export to lithium processing plant (ON/OFF Valve 1510-XV-9530) only when all the following conditions are satisfied.

CONFIGURATION:

Initiator(s):

Permissive from Lithium Processing Plant Ready Indication (XXXX-XX-XXXX).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5195

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I-201, TREATED WATER EXPORT PUMPS 1560-PP-006 AND 1560-PP-007 (STANDBY) START PERMISSIVE ON:

DESCRIPTION:

The interlock permits treated water export pumps 1560-PP-006/007 to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Permissive from Lithium Processing Plant Ready Indication (XXXX-XX-XXXX).
NEED TO HAVE THE LITHIUM PROCESS PLANT READY SIGNAL TAG NUMBER)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5134

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I-202, 150 PSIG STEAM EXPORT STATION (1560-PCV-6011) LITHIUM PROCESSING PLANT START PERMISSIVE ON:

DESCRIPTION:

The interlock permits 150PSIG Steam Station (1560-PCV-6011) to lithium process plant to start only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

Permissive from Lithium Processing Plant Ready Indication (XXXX-XX-XXXX).
NEED TO HAVE THE LITHIUM PROCESS PLANT READY SIGNAL TAG NUMBER)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5165

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I-302, PERMISSIVE TO OPEN BYPASS DESUPERHEATER VALVE 1560-PCV-6672:

DESCRIPTION:

The interlock permits opening bypass desuperheater valve 1560-PCV-6672 only when the following condition is satisfied.

CONFIGURATION:

Permissive(s):

PRESSURE OF EXHAUST STEAM LINE > L (1560-PSL-9304)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5165

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I-307, AIR COOLED CONDENSER (ACC) UNIT FAN START UP PERMISSIVES

DESCRIPTION:

The interlock permits air cooled condenser (ACC) unit fan start only when all the following condition is satisfied.

CONFIGURATION:

Permissive(s):

ACC condenser air take-off Temperature (1560-TIA-9330) > Ambient temperature (1560-TIA-9300) +50°F

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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I-308, PERMISSIVE FOR AIR COOLED CONDENSER (ACC) START-UP SEQUENCE:

DESCRIPTION:

The interlock permits ACC Start-up Sequence only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

High Condensate Tank Level (1560-LSLL-9320)

AND

Fan Assembly in Auto Mode (1560-AN-001/1560-AN-002/1560-AN-003)

AND

{{Condensate Extraction Pump (1560-PP-015) Running

OR

Condensate Extraction Pump (1560-PP-016) Running}}

AND

Gland Steam Condenser Flow NOT Low-Low (1560-FSLL-7511).

AND

Air Removal System is in Auto Mode (1560-XA-XXXX)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174
 1560-PR-PID-0000-EXP-5175
 1560-PR-PID-0000-EXP-5176

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I-317, PERMISSIVE FOR OPENING ACC HOLDING SKID AIR SUCTION VALVE(V1560-9340) ON:

DESCRIPTION:

The interlock permits opening ACC holding skid air suction valve (V1560-9340) only when the following condition is satisfied.

CONFIGURATION:

Permissive(s):

Main Holding Skid Motive Steam Valve (V1560-9363) is open.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

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I-320, PERMISSIVE FOR OPENING EXHAUST STEAM LINE VACUUM BREAKER (V1560-9308) ON:

DESCRIPTION:

The interlock permits opening exhaust steam line vacuum breaker (V1560-9308) only when the following condition is satisfied.

CONFIGURATION:

Permissive(s):

Steam Turbo Generator Speed (1560-XS-6201) < 300 rpm

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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I-323, PERMISSIVE FOR PUT FAN ASSEMBLY FUNCTION GROUP IN AUTO ON:

DESCRIPTION:

The interlock permits putting fan assembly function group in AUTO mode only when all the following conditions are satisfied.

CONFIGURATION:

Permissive(s):

ALL motors (1560-AN001/1560-AN002/156-AN003) available in Auto Mode.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

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16. ON-OFF CONTROL INTERLOCKS

I-63A, OPEN RAW WATER TO OXIDATION BLOWERS AIR SPARGER ON/OFF VALVE (1530-XV-0656):

I-63B, CLOSE RAW WATER TO OXIDATION BLOWERS AIR SPARGER ON/OFF VALVE (1530-XV-0656):

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will open and close the On/Off Valve 1530-XV-0656.

CONFIGURATION:

Initiator(s):

Oxidation Blowers (1530-GB-001 or 1530-GB-002) is Running'

THUS:

On/Off Valve 1530-XV-0656 opens.

OR

Oxidation Blowers (1530-GB-001 or 1530-GB-002) is Not Running

THUS:

On/Off Valve 1530-XV-0656 closes.

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1530-PR-PID-0000-EXP-5106

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I-103, START/STOP LIQUID SULFUR UNLOADING PUMPS 1510-PP-001 AND 1510-PP-002:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will start/stop the liquid sulfur unloading pumps (1510-PP-001 and 1510-PP-002).

CONFIGURATION:

Initiator(s):

- START SULFUR UNLOADING PUMPS 1510-PP-001/1510-PP-002 ON:
 - A. HIGH LEVEL OF SULFUR PIT (1510-LSH-1007A OR 1510-LSH-1007B)
- STOP SULFUR UNLOADING PUMPS 1510-PP-001/1510-PP-002 ON:
 - B. LOW SULFUR UNLOADING PIT LEVEL (1510-LSL-1007A OR 1510-LSL-1007B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5010

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I-117 START / STOP WATER TREATMENT AREA SUMP PUMPS 1560-PP-012 OR 1560-PP-013:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Area Sump Pump 1560-PP-012 or 1560-PP-013.

CONFIGURATION:

Initiator(s):

- ON HIGH-HIGH SUMP LEVEL (1560-LAHH-3601):
START BOTH WATER TREATMENT AREA SUMP PUMPS (1560-PP-012 & 1560-PP-013)
- ON HIGH SUMP LEVEL (1560-LSH-3601):
START ONLY ONE WATER TREATMENT AREA SUMP PUMPS (1560-PP-012 OR 1560-PP-013)
- ON LOW SUMP LEVEL (1560-LSL-3601):
START ONLY ONE WATER TREATMENT AREA SUMP PUMPS (1560-PP-012 OR 1560-PP-013)
- ON LOW-LOW SUMP LEVEL (1560-LALL-3601):
AN ALARM WILL BE INITIATED IN DCS

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5136

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I-119, START BOILER BLOWDOWN SUMP PUMP 1540-PP-001:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Boiler Blowdown Sump Pump 1540-PP-001.

CONFIGURATION:

Initiator(s):

High Boiler Blowdown Sump Level (1540-LSH-7655)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5076

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I-120, STOP BOILER BLOWDOWN SUMP PUMP 1540-PP-001:

DESCRIPTION:

Upon activation of the below initiator, this interlock will stop the Boiler Blowdown Sump Pump 1540-PP-001.

CONFIGURATION:

Initiator(s):

Low Boiler Blowdown Sump Level (1540-LSL-7655)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5076

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I-122, OPEN/CLOSE DEAERATOR OVERFLOW VALVE 1540-XV-6843:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will open the Deaerator Overflow On/Off Valve 1540-XV-6843.

CONFIGURATION:

Initiator(s):

- OPEN DEAERATOR OVERFLOW VALVE 1540-LV-6843 ON:
 - A. HIGH-HIGH DEAERATOR LEVEL (1540-LSHH-6841 & 1540-LSH-6840
SHOULD THIS BE 1002 OR 2002?)
- CLOSE DEAERATOR OVERFLOW VALVE LV-6843
 - B. NORMAL DEAERATOR LEVEL (1540-LSH-6841)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5176

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-137, START AND STOP ACID PLANT COOLING TOWER FAN (1550-CT-001) ON:

DESCRIPTION:

Upon activation of any of the initiators listed below, this interlock will start and stop Cooling Tower Fan (1550-CT-001).

CONFIGURATION:

Initiator(s):

Stops the Fan on Cooling Water Header Temperature Low (1550-TSL-2430),

OR

Starts the Fan on Cooling Water Header Temperature High (1550-TSH-2430).

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1550-PR-PID-0000-EXP-5124

	Interlock Narrative Sulfuric Acid Plant		Revision	A
			Date	2025-11-17
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I-142, SULFUR VAPOR SCRUBBER (1510-PK-007) FLUSH SEQUENCE STARTS ON:

- Close Make-up Water Valve (1510-LCV-1524). Status is confirmed with 1510-XIC-1524,
- Open Scrubber Drain Valve (1510-XEV-1512) for XX seconds,
- Open Water Chase Valve (1510-XEV-1526),
- Close Water Chase Valve (1510-XEV-1526) after Scrubber Drain Valve (1510-XEV-1512) closed. Status is confirmed with 1510-XIC-1512,
- Dis-engage Sulfur Vapor Scrubber Flush Hand Switch (1510-HS-1512).

DESCRIPTION:

Upon activation of the below initiator, above flushing operations will begin.

CONFIGURATION:

Initiator(s):

Sulfur Vapor Scrubber Flush Hand Switch Engaged (1510-HS-1512)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1510-PR-PID-0000-EXP-5015

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-153B, START/STOP ACID HEATER (1570-ZE-003) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Acid Heater 1570-ZE-003.

CONFIGURATION:

Initiator(s):

- STOP ACID HEATER (1570-ZE-003) ON:
 - A. HIGH ACID HEATER 1570-ZE-003 TEMPERATURE (1570-TSH-8806A)
- START ACID HEATER (1570-ZE-003)
 - B. LOW ACID HEATER 1570- ZE-003 TEMPERATURE (1570-TSL-8806A)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

	Interlock Narrative Sulfuric Acid Plant		Revision	A
			Date	2025-11-17
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I-154B, START/STOP ACID HEATER (1570-ZE-003) ON:

DESCRIPTION:

Upon activation of the below initiator, this interlock will start the Acid Heater 1570-ZE-004.

CONFIGURATION:

Initiator(s):

- STOP ACID HEATER (1570-ZE-004) ON:
 - A. HIGH ACID HEATER 1570- ZE-004 TEMPERATURE (1570-TSH-8806B)
- START ACID HEATER (1570- ZE-004)
 - B. LOW ACID HEATER 1570-ZE-004 TEMPERATURE (1570-TSL-8806B)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1570-PR-PID-0000-EXP-5188

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-163, STARTUP BURNER SYSTEM PURGE FORCE ON:

DESCRIPTION:

This interlock will monitor the Startup Burner Purge Force is active and will be monitored by DCS signal 1540-XA-2990.

CONFIGURATION:

Initiator(s):

- I. SET MAIN COMPRESSOR CAPACITY CONTROL (1540-HIC-4030) TO 40% OUTPUT, ENABLES 1540-FS-4070 AND SEND SIGNAL LOGIC "1" TO BMS ONCE FLOWRATE IS REACHED ON:
 - A. PURGE FORCE SIGNAL IS RECEIVED FROM BMS (1540-XI-2990)
- II. SET MAIN COMPRESSOR CAPACITY CONTROL (1540-HIC-4030) TO 0% OUTPUT AND DISABLES FS- 4070 ON:
 - B. PURGE FORCE SIGNAL IS REVOKED FROM BMS (1540-XI-2990)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1540-PR-PID-0000-EXP-5029

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-318, STARTUP BURNER SYSTEM PURGE FORCE ON:

DESCRIPTION:

This interlock will monitor the Startup Burner Purge Force is active and will be monitored by DCS signal 1540-XA-2990.

CONFIGURATION:

Initiator(s):

- OPEN ACC HOLDING UNIT FIRST STAGE EJECTOR (1560-BN-011/012) ON:
A. STEAM/AIR PRESSURE HIGH (1560-PAH-9336)
- CLOSE ACC HOLDING UNIT FIRST STAGE EJECTOR (1560-BN-011/012) ON:
B. STEAM/AIR PRESSURE LOW (1560-PAL-9336)

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5177

	Interlock Narrative Sulfuric Acid Plant		Revision	A
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I-321, CLOSE EXHAUST STEAM LINE VACUUM BREAKER (V1560-9308) ON:

DESCRIPTION:

This interlock will close Exhaust Steam Line Vacuum Breaker (V1560-9308) ON.

CONFIGURATION:

Initiator(s):

STEAM TURBO GENERATOR SPEED (1560-XS-6201) IS LESS THAN 300 RPM

Reference:

Instrument Index: 1500-IN-LST-0000-EXP-5001, Rev. 1

REFERENCES:

P&ID No: 1560-PR-PID-0000-EXP-5174

DOCUMENT END